Paths, combinations and Pascal

High school · counting · Worked solutions · 05

Exercise 1. Find $\binom{8}{2}$ and explain why it equals $\binom{8}{6}$.

Solution. $\binom{8}{2} = 8 \times \frac{7}{2} = 28$. Choosing two excluded objects is equivalent to choosing six included objects.

Exercise 2. Count monotone paths from $(0,0)$ to $(4,3)$.

Solution. Choose four rightward steps among seven: $\binom{7}{4} = 35$.

Exercise 3. How many of these paths pass through $(2,1)$?

Solution. $\binom{3}{2} \times \binom{4}{2} = 3 \times 6 = 18$.

Exercise 4. Choose a team of three from eight people. Two specified people cannot be together. How many teams remain possible?

Solution. There are $\binom{8}{3} = 56$ teams. Exactly 6 contain both specified people. Therefore 50 teams remain.

Problem · Investigate, justify, conclude. Count monotone paths from $(0,0)$ to $(5,4)$ that avoid $A(2,1)$ and $B(3,3)$. Count all paths, then those passing through A, through B and through both. Explain why the last category must be added back.

Solution. Total: $\binom{9}{5} = 126$. Through A: $\binom{3}{2}\binom{6}{3} = 60$. Through B: $\binom{6}{3}\binom{3}{2} = 60$. Through both: $3 \times \binom{3}{1} \times 3 = 27$. Paths through both were subtracted twice; adding them back once gives $126 - 60 - 60 + 27 = 33$.